IBRAHIM ARSHAD

AI Systems Engineer

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Summary

AI systems engineer with 5+ years of experience turning product requirements into production systems, with a focus on reliable agents, evals, infrastructure, distributed systems, and blockchain protocols.

Selected Work

Agent Runtime Reliability Lab

Independent project

- Built an AI-assisted reliability lab investigating agent approval, checkpoint recovery, retries, timeouts, and persistent execution budgets.
- Reproduced duplicate effects after a post-commit crash and tested receiver-side idempotency that preserved one committed action across two tool attempts.
- Implemented receiver reconciliation that detects duplicate actions, checks approved contents, and reports uncertainty when outcome evidence is unavailable.
- Published reproducible experiments, offline evidence validation, and 130 development tests; reproduced five crash/recovery cases in a fresh environment using published packages.

Python · Deep Agents · LangChain · LangGraph · SQLite

SettleDiff

Transaction forensics for agent purchases

- Built an evidence-first system for reconstructing what an agent purchase promised, executed, and actually settled.
- Added deterministic checks across payment execution and settlement evidence instead of relying only on provider receipts.
- Integrated Context.dev and Hyperfusion for external evidence and explanation workflows.

Python · Perflo · x402 · EIP-3009 · Context.dev · Hyperfusion

Vault Steward

Local-first integrity system for Obsidian

- Built and published an Obsidian plugin that combines deterministic checks with bounded model-assisted review.
- Detects broken references, schema issues, contradictions, stale knowledge, orphaned tasks, and unresolved decisions.
- Keeps vault data local and requires explicit user approval before applying proposed edits.

TypeScript · Obsidian · Ollama · OpenAI · agent workflows · evals

Experience

Hypotenuse Labs — Remote

Software Engineer — *Jan 2025 – Present*

- Lead client projects end-to-end, translating stakeholder requirements into technical scope, system architecture, and production delivery.
- Design product and system architecture across backend, AI, and protocol-heavy applications.
- Built a deterministic Rust perpetual futures engine covering order matching, margin, funding, oracle safety, and liquidations.
- Built a Rust zk-rollup prototype for policy-constrained, provable AI workflow execution.
- Built an MCP-based integration gateway for AI providers and tools.
- Built Stellar vault and reward infrastructure plus Merkle-based reward services.
- Deploy and operate production backend infrastructure on AWS.

Lucia Protocol — Remote

Blockchain Engineer — *Apr 2024 – Dec 2024*

- Built backend services and APIs for distributed financial products and transaction processing.
- Collaborated on product architecture and system design from requirements through implementation.
- Integrated backend services with blockchain networks and external systems.

Fantom Foundation — Remote

Blockchain Developer — *Aug 2021 – Apr 2024*

- Built backend infrastructure, indexing pipelines, GraphQL services, and APIs for production blockchain applications.
- Designed systems for high-volume event processing and data consistency.
- Improved backend reliability through fault-tolerant architecture and observability.
- Performed smart-contract security reviews and supported ecosystem production deployments.

Open Source Contributions

- **vLLM** · merged prefill-path reliability fix
- **LangGraphJS** · merged RemoteGraph checkpoint-state fix
- Additional contributions across **CrewAI, Reth, Alloy, Qdrant, and Aida**, covering agent execution, inference, transaction lifecycle, storage, and infrastructure reliability.

Technical

Languages: Rust, Python, TypeScript, Solidity
AI Systems: Agents, evals, MCP, agent harnesses, structured outputs, RAG
Product & Systems: Technical scoping, product architecture, system design, API design
Infrastructure: AWS, Docker, GitHub Actions, CI/CD

Education

BSc Computer Science — 2015–2019

University of Wollongong Dubai